

Building the dataset without a camera

Where the images actually are, how to fetch them, and what the numbers will and will not mean afterwards.

READY-MADE DATASETS

None

CLASSES THAT ARE SPECIES

2, maybe 3

CLASSES THAT ARE CULTIVARS

3, maybe 4

VERIFIED OPEN PHOTOS

~134 total

TOOLS SHIPPED

2, tested

RIG TRANSFER

Do not expect it

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What changes, and what it costs you

I re-ran the search that produced `docs/01-dataset-sourcing-guide.md`, and the answer has not moved: **there is no public fruit-image dataset for your six varieties**. The Philippine mango literature that does exist is leaf-based variety recognition, or Carabao-only size grading. Roboflow's largest mango classification set is 3,441 images of generic mangoes with no variety labels that match yours.

So there is nothing to download in one piece. What follows is the assembly job instead: five sources, ranked by how much you can trust their labels, plus two tools I wrote and tested against the live APIs.

The good news: the pipeline already expects this

You do not need to change any code. `mango_variety/dataset.py` handles single-image fruits deliberately — from the docstring of `fruit_id_for`:

```
DATASET.PY, LINE 33
```

```
"A filename with no view marker is treated as its own single-view fruit, which is the right
behaviour for scraped web images."
```

A file called `carabao_inat144605481_248222867.jpg` has no `_v<n>`, so it becomes one fruit with one view. Grouping still works, the split is still clean, nothing leaks. Drop the files into the same six directories and every script downstream behaves.

The bad news, stated once

Two costs, and you should know both before spending a week on this.

The domain gap. Your rig has fixed lighting, a known conveyor background and five fixed angles. Web images are studio shots, market stalls, hands holding fruit, white-background product photos. A model trained on those learns what a mango *photograph* looks like, not what a mango looks like under your lights. Accuracy on a web-image test split will not transfer to the machine, and the gap is usually large — not a few points.

Label noise. Three of your classes — Carabao, Apple Mango, Indian — are all *Mangifera indica*, separated by fairly subtle differences in shape and blush. Search engines do not know the difference either. Expect a meaningful fraction of any harvested class to be mislabelled at the source, and expect it to be worst exactly where the classes are most similar.

Neither cost makes this pointless. It makes the pipeline demonstrable end to end, which is worth real marks, and it gives you a genuine baseline. It just is not the number you can claim the machine will hit.

IF ANY PHOTOGRAPHY EVER BECOMES POSSIBLE

Twenty mangoes per class shot on a phone against a sheet of paper will beat five hundred scraped images, because they match deployment far better. You have said that is not available, so the rest of this document assumes it is not — but if the constraint ever loosens, even a little photography changes the outcome more than any amount of scraping.

Phase 1 · Know what you are searching for

TWO HOURS · THE HIGHEST-LEVERAGE STEP

Most of the difficulty here is that four of your six class names are Philippine local names that search engines do not index well. Getting the synonyms right is worth more than any scraping tool.

1

The name map

READ BEFORE SEARCHING · DETERMINES EVERYTHING DOWNSTREAM

I looked each of your six up. The single most useful finding: **two of your classes are not *Mangifera indica* at all — they are separate species.** That matters enormously, because separate species have scientific names, and scientific names are indexed by biodiversity databases with expert-verified labels. Cultivars of one species are not.

CLASS	WHAT IT ACTUALLY IS	ALSO CALLED	LABEL SOURCE
Wani	<i>Mangifera caesia</i> species	binjai, white mango, bayuno, mangga wani	iNaturalist, GBIF — verified
Kabayo	<i>Mangifera foetida</i> (likely) species	horse mango, manggang kabayo, bachang, machang, limus, mamut	iNaturalist, GBIF — verified , but confirm the equivalence
Indian	<i>M. indica</i> 'Katchamitha' cultivar	Katchamitha, Katchamita, kachamita	Wikipedia-documented; search by cultivar name
Chupadera	<i>M. indica</i> — possibly the Pico cultivar unconfirmed	possibly pico, piko, padero, supsupin, pajo, bioko	Conflicting sources — see the warning below
Carabao	<i>M. indica</i> 'Carabao' cultivar	manggang kalabaw, Philippine mango, Manila mango	Heavily documented; easiest class to harvest
Apple Mango	<i>M. indica</i> , round with red blush cultivar group	mangga apple; likely Kenyan-introduced	Ambiguous name — highest label risk

VERIFY CHUPADERA AND KABAYO WITH YOUR ADVISER

A Philippine food publication lists *chupadera* as a regional name for the **Pico** mango. Wikipedia's Pico article does **not** list it, giving only *piko* and *padero*. One of those sources is wrong and I cannot tell which from here.

Same for Kabayo: "horse mango" is the standard English name for *Mangifera foetida*, and Filipino growers use "manggang kabayo" for a long, large native mango — but I could not confirm they are the same fruit.

Ask before you harvest. If Kabayo really is *M. foetida*, Step 3 gives you a properly labelled class for free. If it is a local *M. indica* landrace, those images are a different fruit and will poison the class.

2

The four traps

EACH ONE SILENTLY CORRUPTS A CLASS

SEARCH TERM	WHAT YOU WILL ACTUALLY GET
champagne mango	Ataulfo , a Mexican cultivar. "Champagne" is a US trade name for it, not for Carabao. Never use this term for your Carabao class.
Indian mango	Alphonso, Kesar, Banganapalli and every other Indian cultivar. Your "Indian" class is <i>Katchamitha</i> , a Philippine cultivar that merely carries that nickname. Search the cultivar name, not the nickname.
apple mango	At least three unrelated things — a Kenyan cultivar, a Thai cultivar, and any round mango a seller felt like calling this. Highest-risk class; curate it hardest.
pahutan	<i>Mangifera altissima</i> , a third species entirely. If you follow the Chupadera-equals-Pico trail, do not let pahutan images in with it.

There is a fifth, structural trap worth naming: **a term that returns beautiful, consistent results is suspicious**. Stock-photo sites dominate generic queries, and a class made of stock photos teaches the model to recognise studio lighting. Prefer messier, more local results.

Phase 2 · Harvest, best labels first

TWO TO FOUR DAYS

Work down the tiers in order. Tier 1 gives you a small number of images whose labels an expert has confirmed; by Tier 4 you are curating everything by hand. Fill the trustworthy sources first so you know how big the gap is before you start guessing.

3 Tier 1 — iNaturalist

30 MINUTES · VERIFIED LABELS · SMALL YIELD

Every research-grade iNaturalist observation has been identified by two or more people. For the two species classes this is the only source in this document whose labels you can simply trust.

I wrote `tools/fetch_inaturalist.py` for this and tested it against the live API. Stdlib only, so it runs on your Python 3.14 with no installs:

```
cd "C:\Users\ANGELO GALLEROS\GitHub\MangoScan\MyTasks"

python tools/fetch_inaturalist.py --taxon-id 359080 \
    --label Wani --out data/raw/Wani --max 300 --all-grades

python tools/fetch_inaturalist.py --taxon-id 348333 \
    --label Kabayo --out data/raw/Kabayo --max 300 --all-grades
```

It filters to open licences (CC0 and the CC-BY family), skips all-rights-reserved photos, downloads at ~1024px, rate-limits itself, and writes an `ATtribution_<class>.csv` next to the images.

WHAT YOU WILL ACTUALLY GET — MEASURED, NOT ESTIMATED

CLASS	RESEARCH-GRADE	ALL GRADES
Wani — <i>M. caesia</i>	3 obs / 9 photos	25 obs / 40 photos
Kabayo — <i>M. foetida</i>	7 obs / 16 photos	54 obs / 94 photos

THIS IS BELOW THE FLOOR, AND IT GETS WORSE

134 photos across two classes, against a 60-fruit-per-class minimum. And these are *observations of plants* — a large share will be leaves, bark, flowers or whole trees, not fruit. After culling non-fruit images in Step 8, expect roughly a third to survive.

Use `--all-grades`, as above. It trades some label certainty for four times the volume, and at these counts you need the volume. Then check every image by eye — at this scale that is entirely feasible.

Keep the attribution CSV. CC-BY and CC-BY-NC both require credit, and an appendix listing your image sources is exactly the kind of thing a panel asks for.

4 Tier 2 — GBIF

1 HOUR · TEN TIMES THE VOLUME · MUCH DIRTIER

GBIF aggregates iNaturalist plus herbarium collections and national biodiversity databases. The pools are far bigger:

SPECIES	GBIF KEY	OCCURRENCES WITH IMAGES
<i>Mangifera caesia</i>	3190637	356
<i>Mangifera foetida</i>	3190640	411

MOST OF THESE ARE HERBARIUM SHEETS

A herbarium specimen is a pressed, dried, flattened plant on a mounting card with a label. It is scientifically valuable and completely useless for training a fruit classifier. The jump from 40 photos to 356 is mostly these.

Filter hard, and check by eye. If a "mango" is grey-brown and flat against a white card with a typed label in the corner, that is a herbarium sheet.

GETTING THE IMAGES

1. Go to `gbif.org/occurrence/search`
2. Filter: **Scientific name** = `Mangifera caesia`
3. Filter: **Media type** = `StillImage`
4. Filter: **Basis of record** = `Human observation` — this is the important one. It excludes `PreservedSpecimen`, which is what herbarium sheets are.
5. **Download** → **Simple**. You get a CSV; the image URLs are in the `identifier` column of the associated multimedia file.
6. Register a free GBIF account first — downloads require one, and it gives you a citable DOI for the exact query you ran. Put that DOI in your paper.

With the Human-observation filter applied you are largely back to the iNaturalist set plus a modest number of extra field photos. That is the honest ceiling for these two classes.

5

Tier 3 — Wikimedia Commons

20 MINUTES · TINY BUT CLEAN

Small, well-labelled, and licensed for reuse by definition. The *Mangifera caesia* category holds 14 files — I checked. Not a dataset, but 14 clean images is a real contribution when a class only has 40 candidates.

Categories worth pulling:

- `Category:Mangifera caesia` — Wani
- `Category:Mangifera foetida` — Kabayo
- `Category:Mangifera indica` and its subcategories — general
- `Category:Mangoes of the Philippines`, if present

The category listing is a plain JSON endpoint, so you can see what is there before downloading anything:

```
https://commons.wikimedia.org/w/api.php?action=query  
&list=categorymembers&cmtitle=Category:Mangifera_caesia  
&cmtype=file&cmlimit=500&format=json
```

Paste that into a browser as one line. For a handful of files, downloading by hand from the category page is faster than automating it.

6

Tier 4 — search engines

TWO TO THREE DAYS · THE ONLY OPTION FOR THE CULTIVARS · CURATE EVERYTHING

Carabao, Apple Mango, Indian and probably Chupadera have no biodiversity-database route, because they are cultivars of a species that also covers every other mango on earth. Image search is what is left.

THE BROWSER-EXTENSION ROUTE — RECOMMENDED

For a few hundred images per class, a bulk-image-download browser extension beats any scraping library: you see every thumbnail before it lands on disk, which is most of the curation done for free. Search, scroll until results stop being relevant, then bulk-download the page.

Set the extension's minimum image size to around **400×400**. That one setting removes most thumbnails, icons and logos before they reach you.

THE SCRIPTED ROUTE

If you want it automated, `icrawler` is the maintained option and drives Google, Bing and Flickr from Python:

```
# Use the 3.12 venv from the training handbook - some of these
# dependencies have no Python 3.14 wheels yet.
py -3.12 -m venv .venv312
.venv312\Scripts\activate
pip install icrawler

python -c "
from icrawler.builtin import BingImageCrawler
c = BingImageCrawler(storage={'root_dir': 'data/raw/Carabao'})
c.crawl(keyword='carabao mango fruit', max_num=200,
        min_size=(400, 400), file_idx_offset='auto')
"
```

Run it once per query term, using `file_idx_offset='auto'` so later runs do not overwrite earlier ones. The full query list is in Reference B.

RENAME BEFORE YOU MIX QUERIES

Crawlers write `000001.jpg`, `000002.jpg`. Rename each batch to something traceable — `carabao_bing01_000001.jpg` — so that when one query turns out to have been poisoned you can remove exactly that batch instead of re-curating the whole class.

SEARCH IN FILIPINO TOO

English queries return export-market and stock photography. Tagalog and Cebuano queries return photos taken by people who actually grow and sell the fruit — messier, more varied, and much closer to what your rig will see. Both language sets are in Reference B, and the local-language results are usually the more valuable half.

7

Tier 5 — sellers and institutions

HALF A DAY, THEN WAITING · HIGHEST UPSIDE

MARKETPLACE LISTINGS

Lazada, Shopee and Facebook Marketplace listings for mango seedlings and fruit are photographed by growers, variety by variety, often on plain surfaces in ordinary light. For the obscure classes — Kabayo, Chupadera — a seller listing may be the only labelled photo in existence. Nursery and seedling listings are especially good, because a seller has a strong reason to name the variety correctly.

Watch for duplicates: the same listing photo is reposted across dozens of shops. Step 8 catches exact copies.

ASK THE INSTITUTIONS — DO THIS ON DAY ONE

This is the highest-value item in the whole document, and it costs three emails. Several Philippine institutions hold mango variety image collections that were never published as datasets:

- **PCAARRD (DOST)** — funds Philippine mango research and maintains project archives
- **IPB, UP Los Baños** — Institute of Plant Breeding; holds germplasm collections of Philippine mango varieties
- **DA-Bureau of Plant Industry** — variety registration, with reference photographs
- The **National Mango Research and Development Center**, and any state university with a mango program near you

Write as a student, name your adviser and institution, say exactly what you need — fruit photographs of six named varieties, for a thesis classifier — and offer to cite them. A researcher with a folder of variety photographs from a past project can hand you a better dataset than weeks of scraping, and they are often glad the images get used. Send these before you start Tier 4; replies take time, and a positive one changes your whole plan.

Phase 3 - Curate ONE TO TWO DAYS · DO NOT SKIP

Harvested data is perhaps half usable. This phase is where a scraped set either becomes a dataset or stays a folder of pictures.

8 Dedupe, cull, and the leak you cannot see

RUN AFTER EVERY HARVEST · TESTED TOOL

WHY DUPLICATES ARE THE SAME DISASTER AS BEFORE

The training handbook is built around one failure: an image in test that the model already saw in training. Scraping reintroduces it by a different route. The same seller photo appears on eight sites; you download all eight; they get different filenames, so the pipeline counts them as **eight different fruits** and scatters them across train and test.

The fruit-grouping logic cannot save you here — as far as it can tell, they genuinely are different fruits. Your accuracy climbs, and every metric looks fine. Again.

`tools/dedupe_images.py` handles it. Stdlib only, defaults to a dry run, and I have tested it against a planted duplicate and a fake image:

```
python tools/dedupe_images.py data/raw/Carabao
python tools/dedupe_images.py data/raw/Carabao --apply
```

```
data/raw/Carabao
  scanned          8
  not images       1
  too small        0 (< 8000 bytes)
  exact duplicates 1
  would keep       6

duplicate groups (keeping the first of each):
  reposted_elsewhere.jpg
    = wani_inat144605481_248222867.jpeg
```

It removes three kinds of junk:

- **Exact duplicates** — byte-identical re-hosts, by SHA-256
- **Files that are not images** — search engines serve HTML error pages with `.jpg` names surprisingly often
- **Files under 8 KB** — thumbnails and icons that carry no detail at 224px

IT CATCHES EXACT COPIES, NOT NEAR-COPIES

A re-saved, resized or watermarked copy has different bytes and will survive. For those you need perceptual hashing, which needs Pillow — in your 3.12 venv, `pip install imagehash pillow` and compare average hashes with a Hamming distance under about 5.

At a few hundred images per class, sorting the folder by thumbnail in Explorer and deleting visible repeats by eye is faster and catches more.

Delete, do not keep:

- Any image that is not a mango fruit — leaves, bark, whole trees, flowers, herbarium sheets, seedlings in pots
- Cartoons, logos, infographics, and text-over-photo marketing images
- Cut fruit, juice, dried mango, and cooked dishes
- Piles and crates where no single fruit is clear
- Anything you are not reasonably confident is the right variety — **when unsure, delete**. One wrong image in a 60-image class is nearly 2% label noise.

Keep a per-class tally as you go. It replaces the label sheet from the camera workflow, and Step 9 needs it.

9

Verify before training

2 MINUTES · SAME COMMAND AS THE HANDBOOK

```
python scripts/01_prepare_dataset.py --config configs/dataset.toml --dry-run
```

The output reads differently now:

```
Carabao          180 images   180 fruits   train/val/test = 126/27/27
```

HERE, IMAGES == FRUITS IS CORRECT

In the camera workflow that equality was *the* alarm — it meant the `_v<n>` markers had failed. With scraped data every image genuinely is its own single-view fruit, so the numbers matching is expected and healthy.

The check that replaces it: **does the fruits count match your curation tally?** If you kept 180 Carabao images and the script says 174, six files did not land where you think they did.

Class balance matters more than before, because harvest yields vary wildly — Carabao will be easy and Chupadera will not. If your largest class is more than about twice your smallest, either harvest more of the small one or cap the large one. Check `macro_avg` against `weighted_avg` in the report afterwards; a wide gap is the imbalance showing up in the metrics.

10

What changes in the metrics

READ BEFORE YOU WRITE THE RESULTS CHAPTER

Training itself is identical — Steps 6 through 9 of the training handbook, no changes. Three numbers now mean something different.

METRIC	WHAT IT MEANS NOW
<code>multiview_gain</code>	Becomes meaningless — expect ~0. It measures the benefit of averaging five views, and every fruit now has one view, so multi-view accuracy equals per-image accuracy by construction. This is not a bug and not the leak warning from the handbook. You simply cannot report this result on scraped data.
<code>multiview_top1_accuracy</code>	Still the headline, but it is now accuracy on <i>web images</i> . Label it that way everywhere. It is not a prediction of rig performance.
<code>multiview_ece</code>	Matters more than ever. Model 5 gates routing on Model 1's confidence, and a model trained on one domain and run on another is usually <i>overconfident</i> exactly where it is wrong. Expect ECE above 0.15 and retune <code>thresholds.variety</code> accordingly.

One thing does not change: `05_predict_multiview.py` still takes five rig images at inference time and averages them. Training on single images does not stop the deployed system from using all five cameras — the averaging happens at prediction time regardless. You just cannot *measure* the benefit until you have multi-view test data.

WORTH DOING: A SECOND TEST SET

If even a handful of rig photographs ever become available — twenty mangoes, four per class — do not add them to training. Hold them out as a separate test set and report both numbers: accuracy on web images, and accuracy on rig images. The gap between those two is a genuine, publishable finding about domain shift, and it is far more interesting than a single inflated accuracy figure.

11

What to write in the paper

PROTECTS THE WORK

State the sourcing constraint plainly in your methodology. A panel that discovers a domain gap you did not mention will treat the whole chapter as unreliable; a panel told about it up front usually treats it as a limitation properly handled. Cover:

- That no public dataset exists for these six varieties, and that you verified this across Kaggle, Mendeley, Roboflow, GBIF and the literature
- Where each class came from, with counts per source — the attribution CSV and the GBIF download DOI give you this for free
- That training images are web-sourced while deployment is a fixed rig, and that reported accuracy is therefore an upper bound on a different distribution, not an estimate of machine performance

- That `multiview_gain` is not evaluable on this data, and why
- Your curation protocol — dedupe, the cull rules, and how many images each class lost. Reviewers trust datasets whose losses are documented

Then recommend rig capture as the clear next step, with the numbers to justify it. Framed that way, this phase is a working end-to-end pipeline plus a measured argument for what the project needs next — which is a legitimate result, and an honest one.

A IDs, endpoints and commands

THING	VALUE
iNaturalist taxon — <i>M. caesia</i> (Wani)	359080
iNaturalist taxon — <i>M. foetida</i> (Kabayo)	348333
GBIF key — <i>M. caesia</i>	3190637
GBIF key — <i>M. foetida</i>	3190640
iNaturalist API	api.inaturalist.org/v1/observations
GBIF search	gbif.org/occurrence/search
Commons API	commons.wikimedia.org/w/api.php

CHECK A POOL SIZE WITHOUT DOWNLOADING

```
python tools/fetch_inaturalist.py --taxon-id 359080 \
  --label Wani --out data/raw/Wani --max 2000 --dry-run --all-grades
```

FULL HARVEST, BOTH SPECIES CLASSES

```
cd "C:\Users\ANGELO GALLEROS\GitHub\MangoScan\MyTasks"

python tools/fetch_inaturalist.py --taxon-id 359080 \
  --label Wani --out data/raw/Wani --max 300 --all-grades
python tools/fetch_inaturalist.py --taxon-id 348333 \
  --label Kabayo --out data/raw/Kabayo --max 300 --all-grades

python tools/dedupe_images.py data/raw/Wani --apply
python tools/dedupe_images.py data/raw/Kabayo --apply

python scripts/01_prepare_dataset.py --config configs/dataset.toml --dry-run
```

Query sheet

Run each term separately and rename the batch before running the next. Filipino terms are listed second and are usually the more valuable half.

CLASS	QUERY TERMS
Carabao	carabao mango fruit · manggang kalabaw · Philippine mango fruit · Manila mango fruit · mangga carabao prutas Never: champagne mango
Apple Mango	apple mango fruit · apple mango Philippines · mangga apple · red blush mango fruit Highest label risk — curate hardest
Indian	katchamitha mango · katchamita mangga · kachamita mango Philippines Never the bare term "Indian mango"
Chupadera	chupadera mango · mangga chupadera If your adviser confirms Pico: pico mango fruit · piko mangga · padero mango Never: pahutan (different species)
Wani	Mangifera caesia · binjai fruit · buah wani · mangga wani · bayuno prutas · white mango binjai
Kabayo	mangga kabayo · manggang kabayo · horse mango fruit · Mangifera foetida · bachang buah · machang fruit

Sources checked 25 August 2026. iNaturalist and GBIF counts were measured by querying their APIs directly while writing this; the Wikimedia count is from the categorymembers endpoint. Variety synonyms come from Wikipedia's Carabao, Pico, Katchamitha and Mangifera caesia / foetida articles, plus Philippine food and horticulture publications – the Chupadera and Kabayo identifications are flagged as unconfirmed for that reason. Both tools were run against live endpoints before being documented here.